



Highly accurate protein structure prediction with AlphaFold

Presenter:

- Jahid Hasan (1024052019)

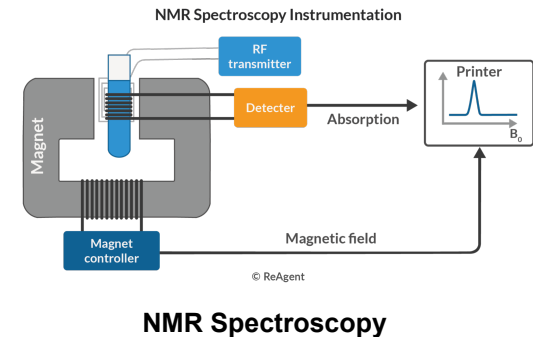
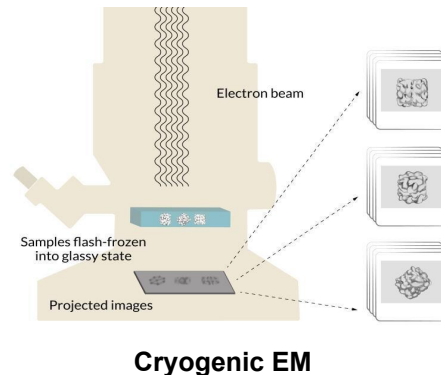
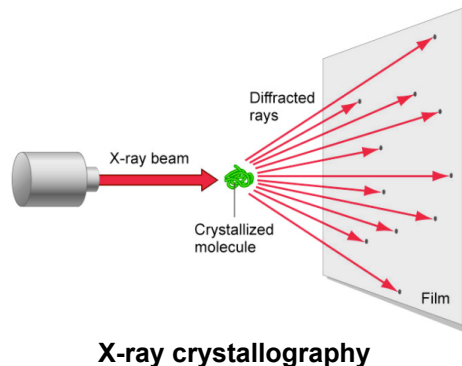
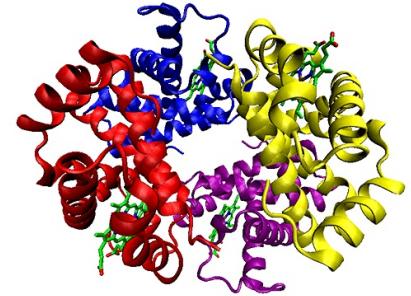


- Background & Motivation
- AlphaFold
- Error Calculation
- Engineering Tricks



Proteins

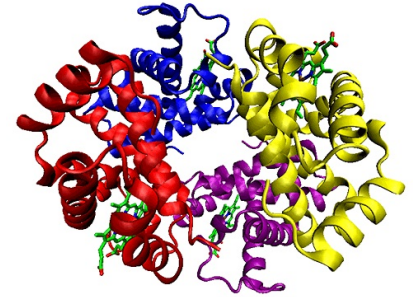
- Essential for life, and their function depends on its 3D structure
- Experimental determination of protein structure is **Slow & Costly**
 - Resulting in structures for only a small fraction of known protein sequences





Predicting 3D structures of Protein

- Open challenge for over 50 years
- Existing methods
 - Physical (Physics-Based) Approaches
 - Evolutionary (Bioinformatics-Based) Approaches

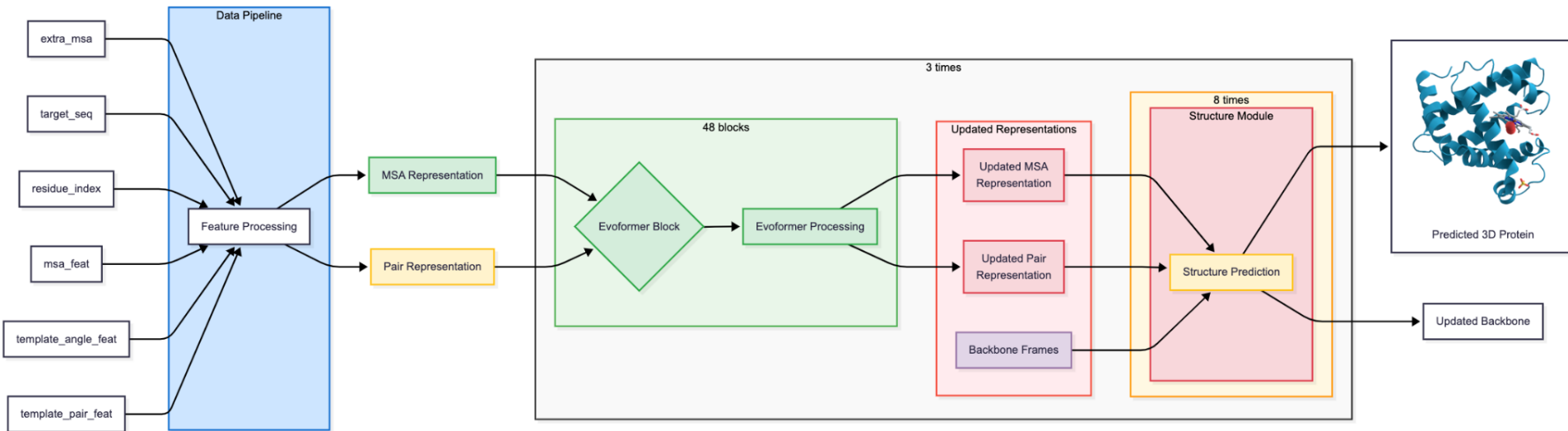




CASP

- In 1994, John Mault started a competition called CASP (Critical Assessment of Structure Prediction)
- Goal is to design a computer model that take a sequence of amino acid output a structure
- Perfect Prediction 100% accuracy, 90% and above is fine enough model.
- In 2018 CASP13, A7D achieve ~68% accuracy , Top one
- In 2020 CASP14 , AlphaFold2 achieved ~87%

The AlphaFold





Data Pipeline

The Data Pipeline is the "Pre-processing" stage

INPUTS OF ALPHAFOLD

extra_msa_feat
(s_e, r, f_e)

residue_index
(r)

target_feat
(r, f)

msa_feat
(s_c, r, f_c)

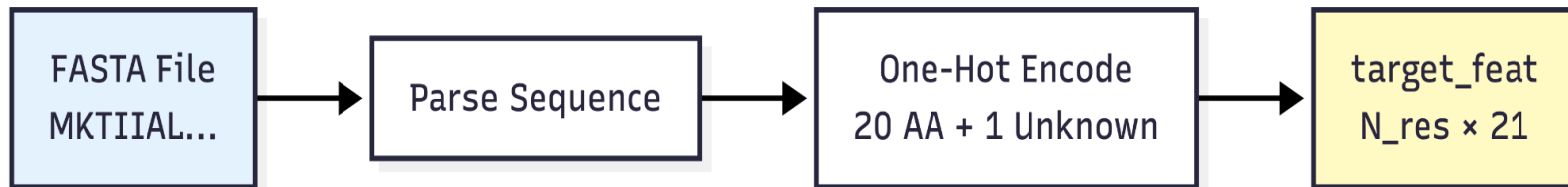
template_angle_feat
(s_t, r, f_a)

template_pair_feat
(s_t, r, r, f_p)



Data Pipeline

target_feat



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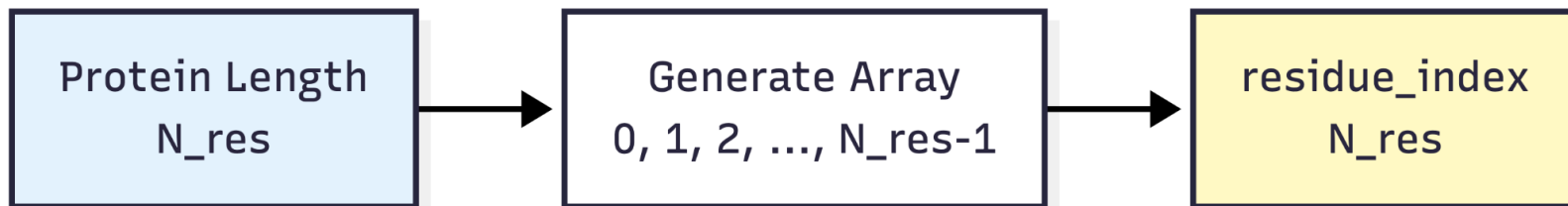
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The AlphaFold



Data Pipeline

residue_index



Residue index :	1	2	3	4	5	6	7	8	9	10
Sequence :	A	B	C	D	E	F	G	H	I	J
Exp. 3D Shape :	Y	Y	Y	N	N	Y	Y	Y	Y	Y
pLDDT Score :	H	H	H	L	L	H	H	H	H	H

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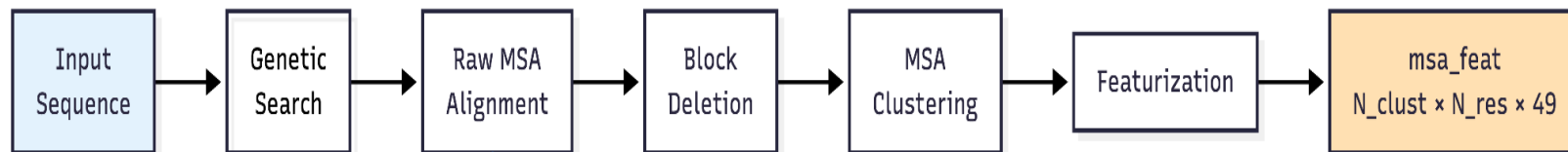
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The AlphaFold



Data Pipeline

msa_feat



- FASTA

- JackHMMER
- HHBlits
- MGnify

Pos: 1 2 3 4 5 ... 256
Seq 1: M K T I A ... L
Seq 2: M K T I A ... L
Seq 3: M K T V A ... L
Seq 4: M R T I A ... L
...
Seq 100k: M K S I A ... L

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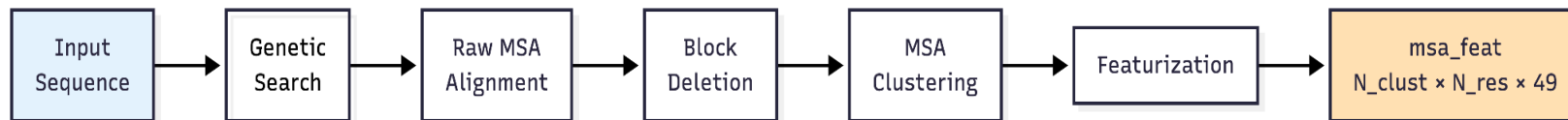
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Data Pipeline

msa_feat



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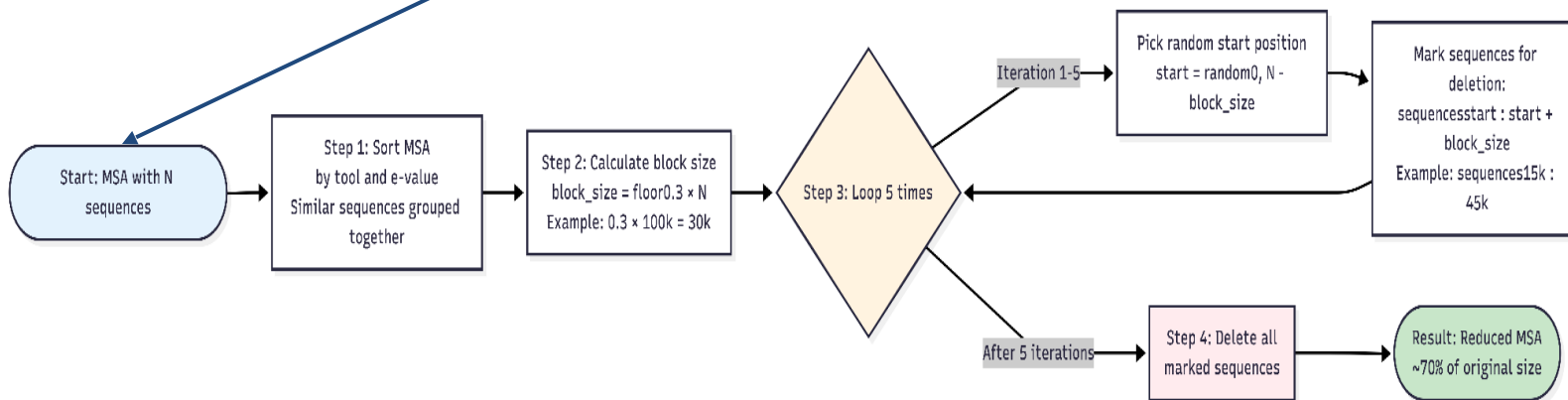
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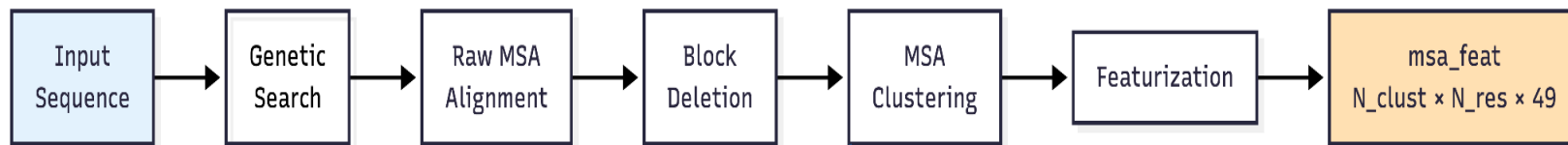


The AlphaFold



Data Pipeline

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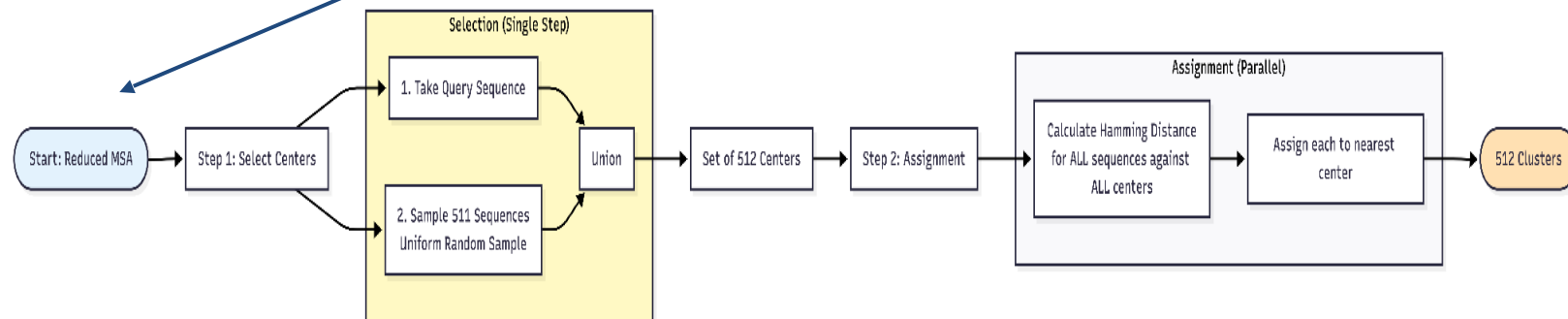


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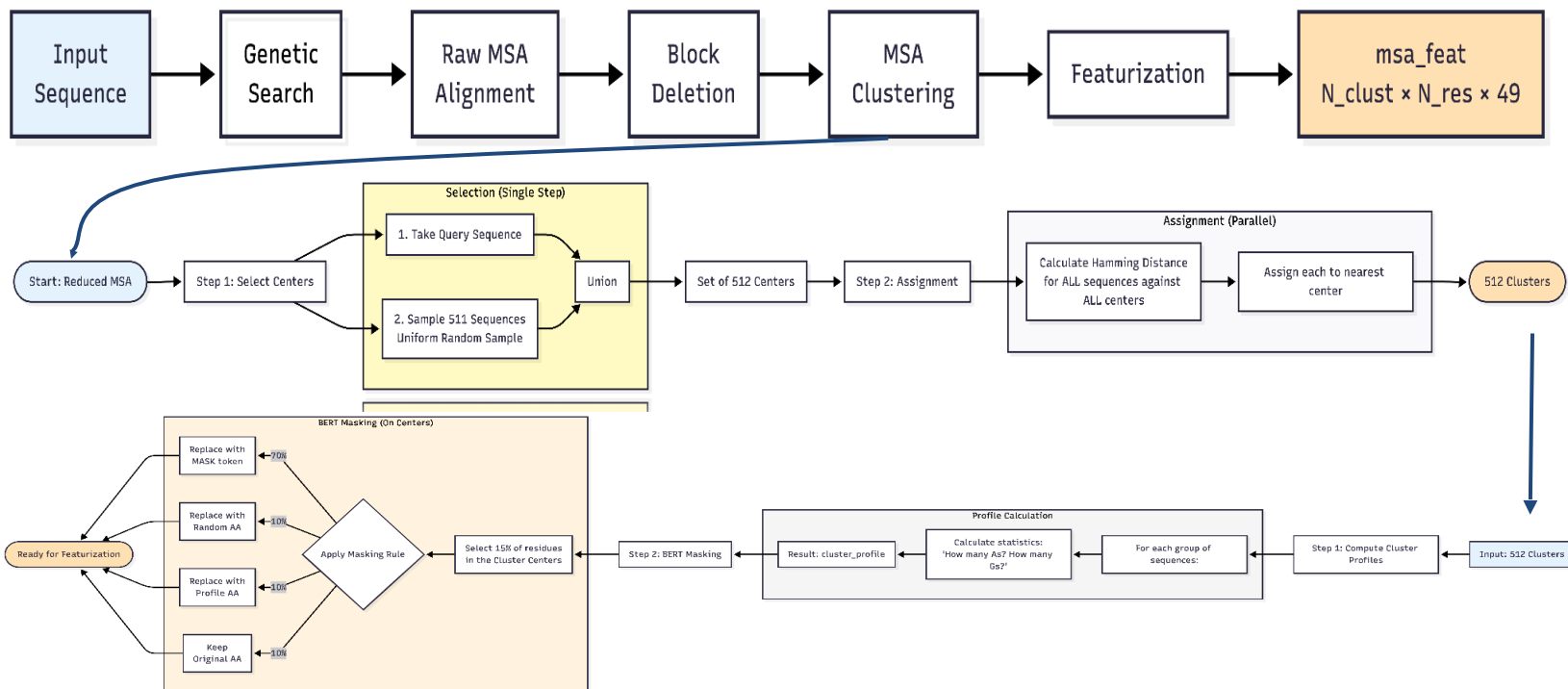
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The AlphaFold



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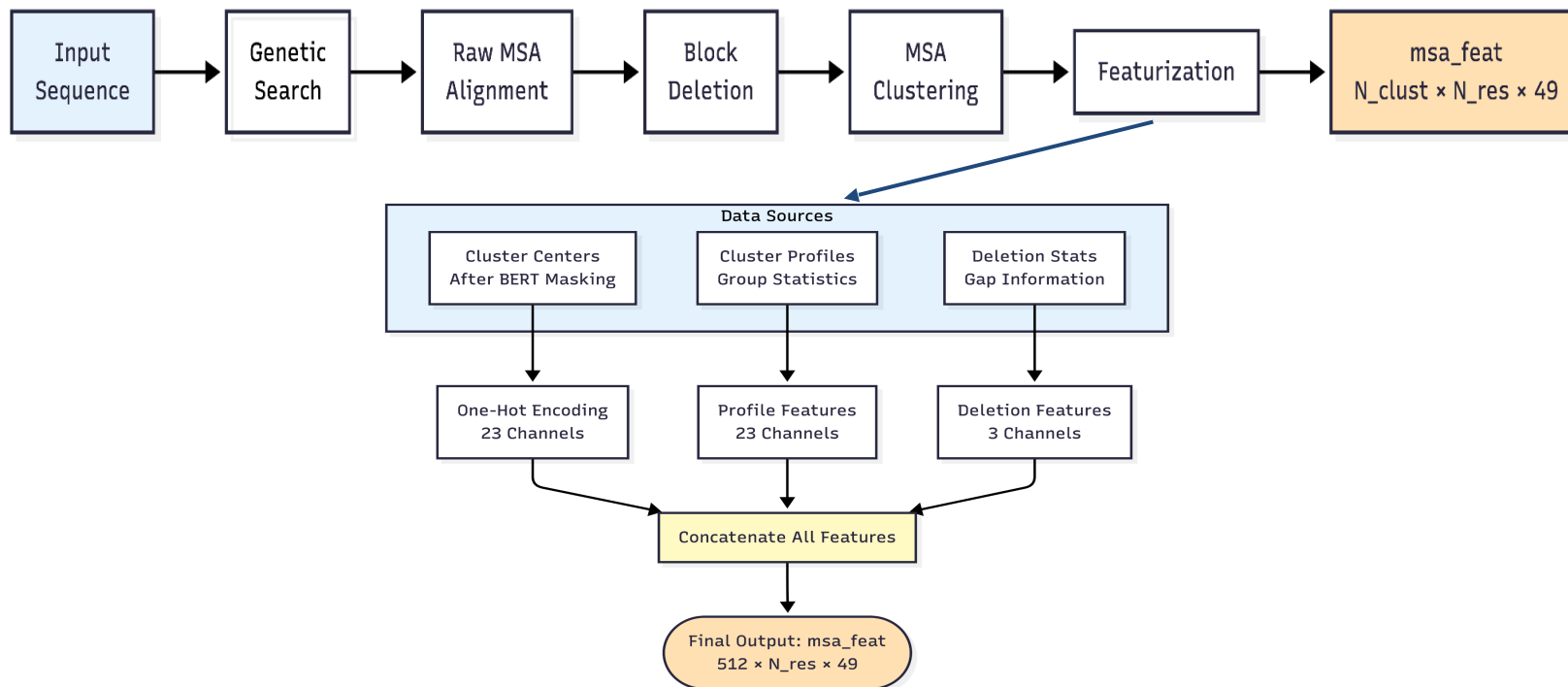
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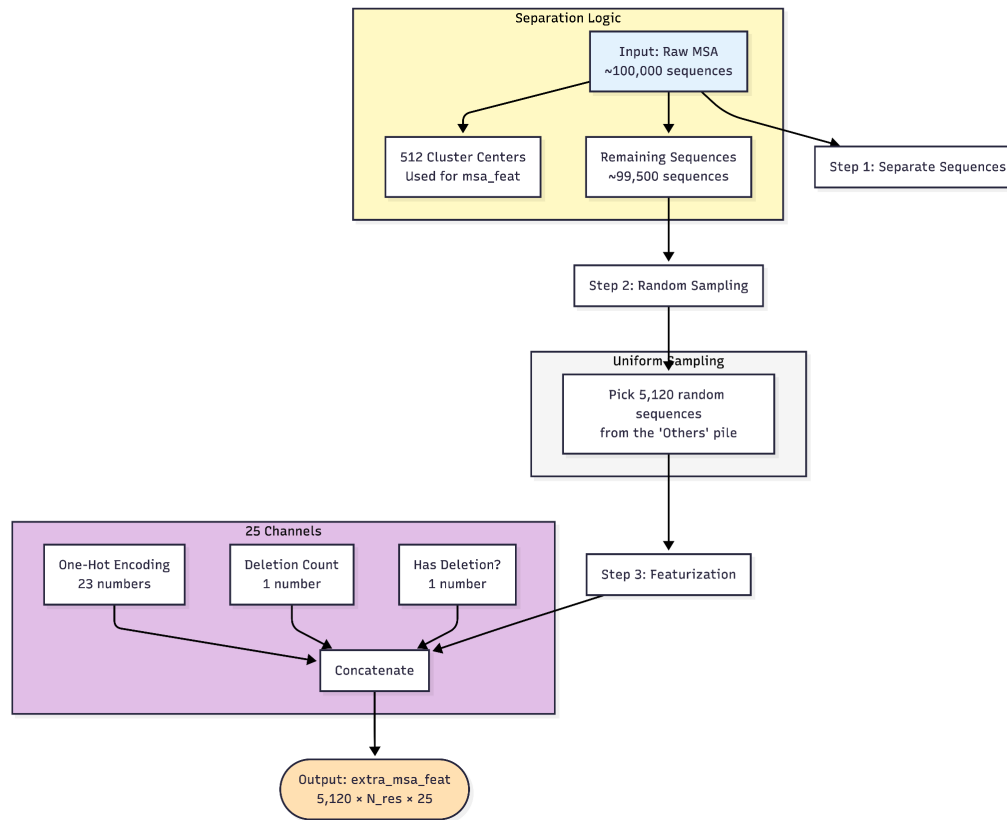
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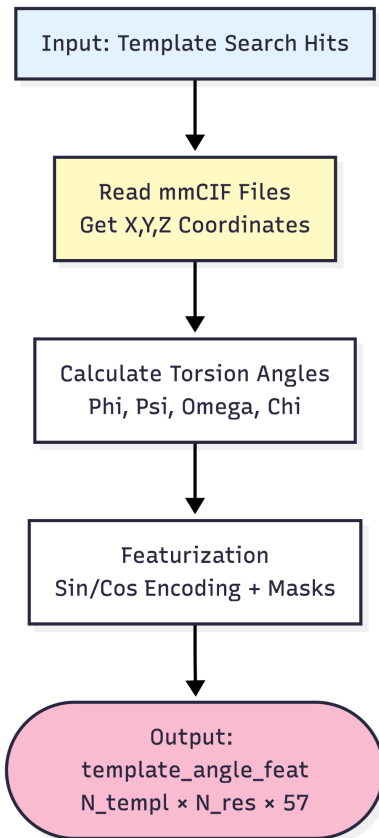
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The AlphaFold



Data Pipeline

template_angle_feat



- Phi (ϕ): Twist between N and C α atoms.
- Psi (ψ): Twist between C α and C atoms.
- Omega (ω): Twist between C and N atoms
- Chi (χ): Twists in the sidechain (C α -C β , etc.)

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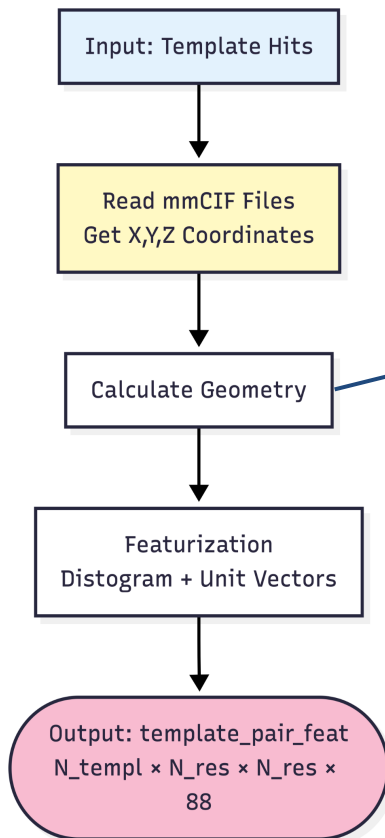
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The AlphaFold



Data Pipeline

template_pair_feat



Pairwise Calculations

- Distances: $\text{C}\alpha$ - $\text{C}\alpha$ distance
- Vectors: Direction between atoms
- Frames: Relative Orientation

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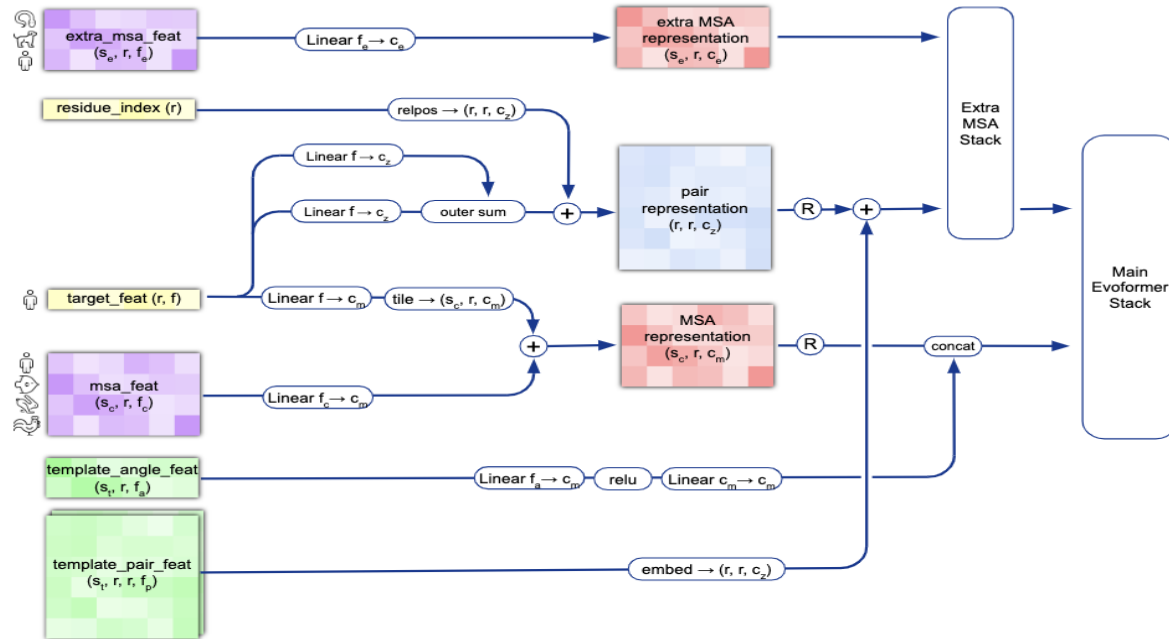
target_feat
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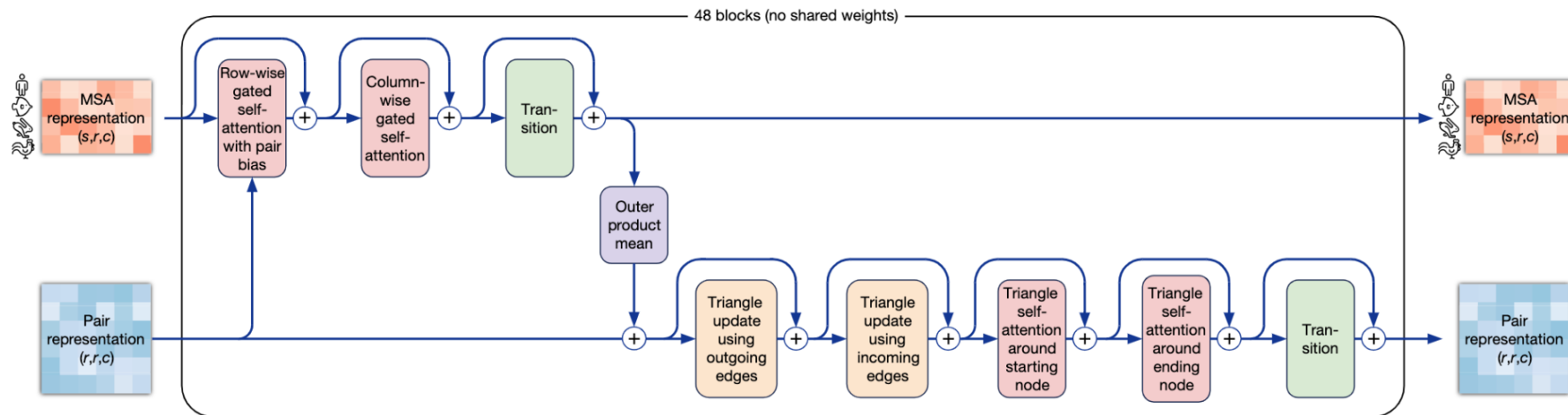
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Data Pipeline





Evoformer

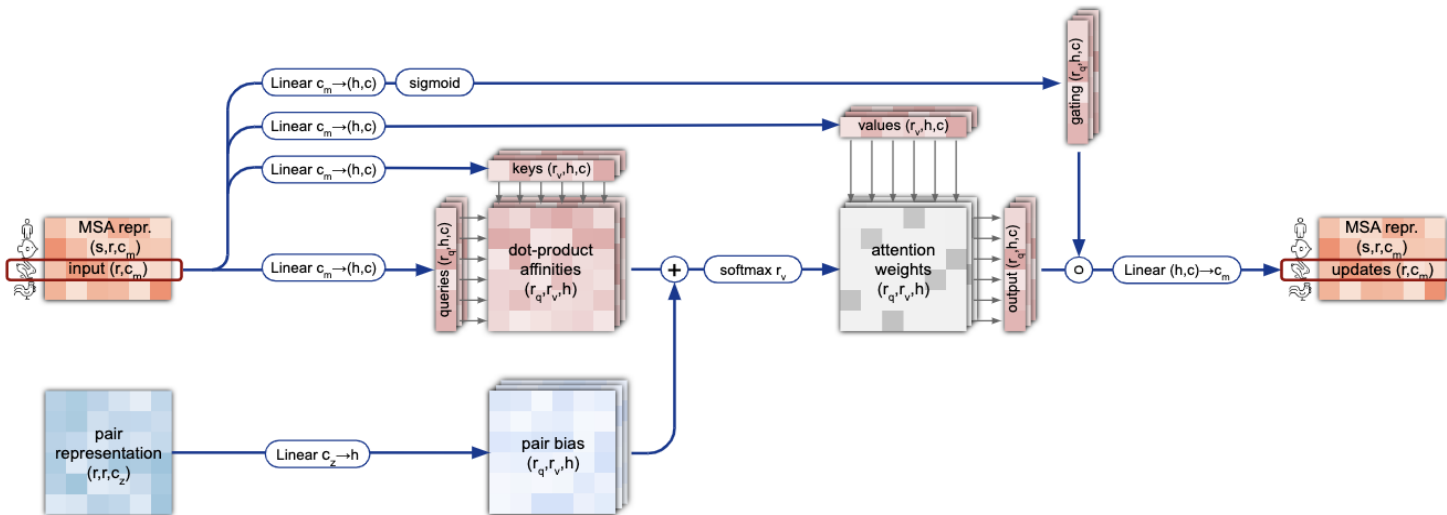
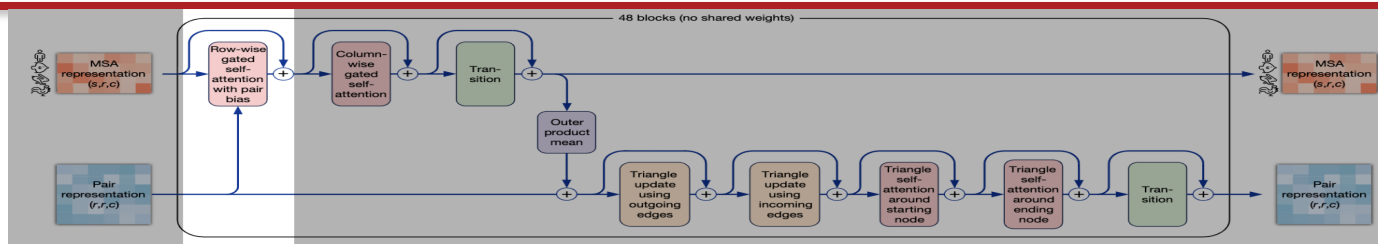


The AlphaFold



Evoformer

Row-wise gated self-attention

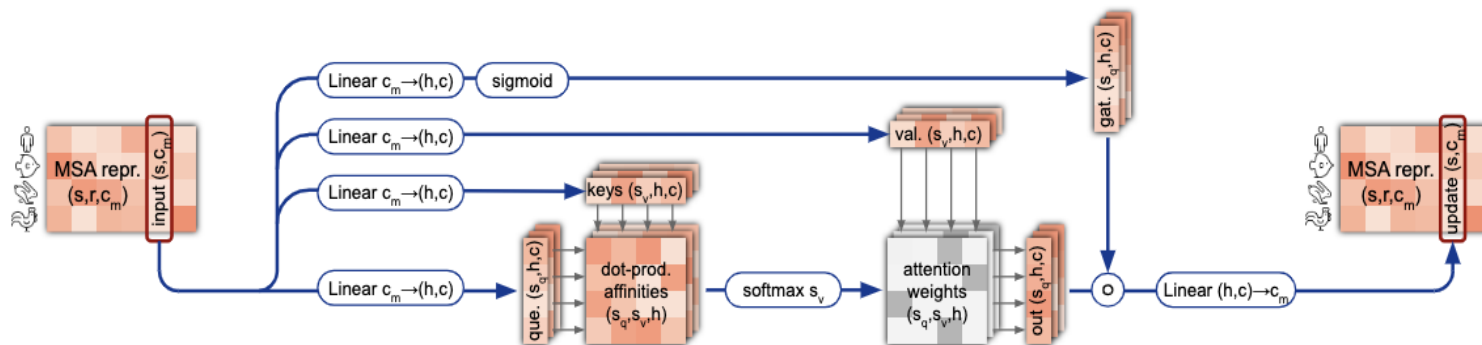
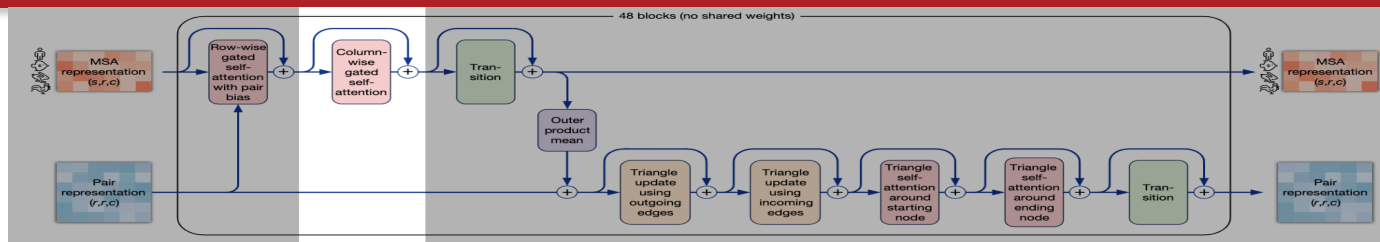


The AlphaFold



Evoformer

Column-wise gated self-attention

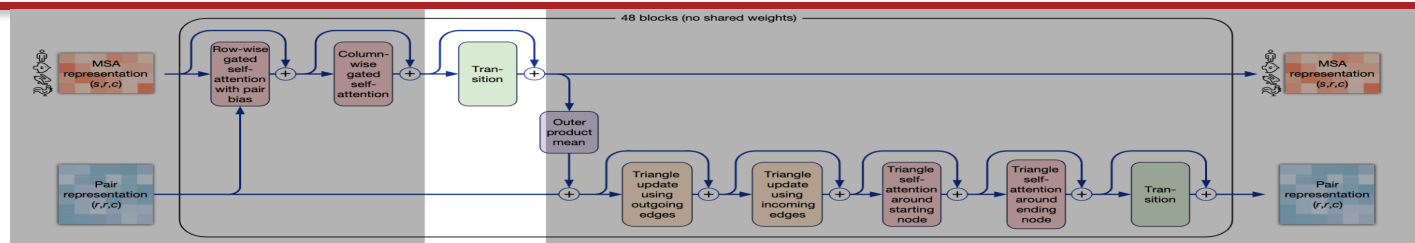


The AlphaFold



Evoformer

MSA transition

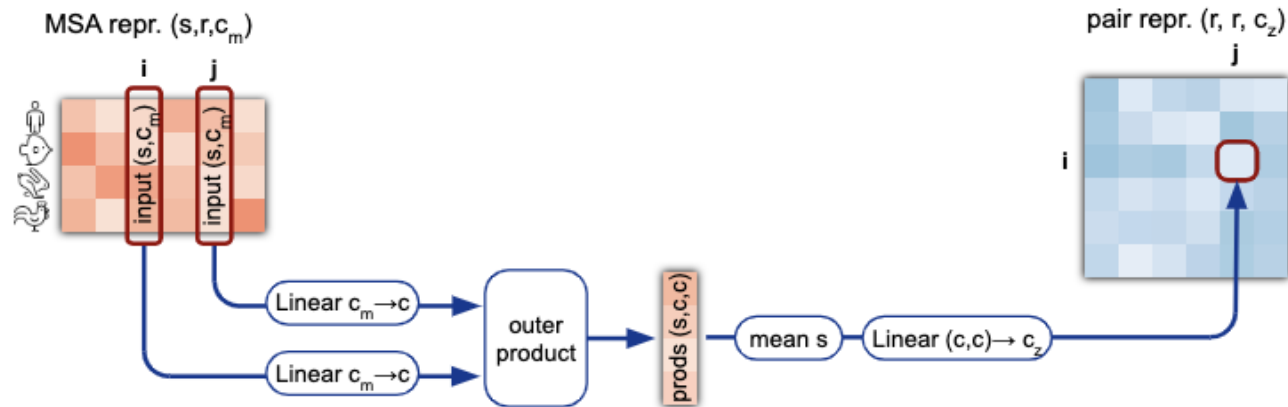
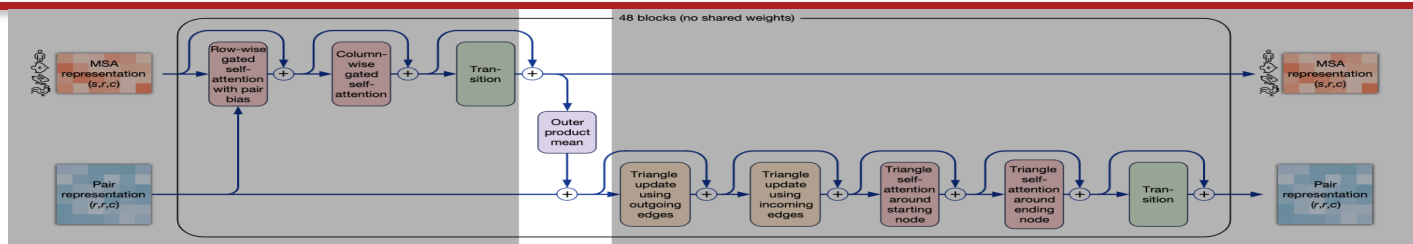


The AlphaFold



Evoformer

Outer product mean

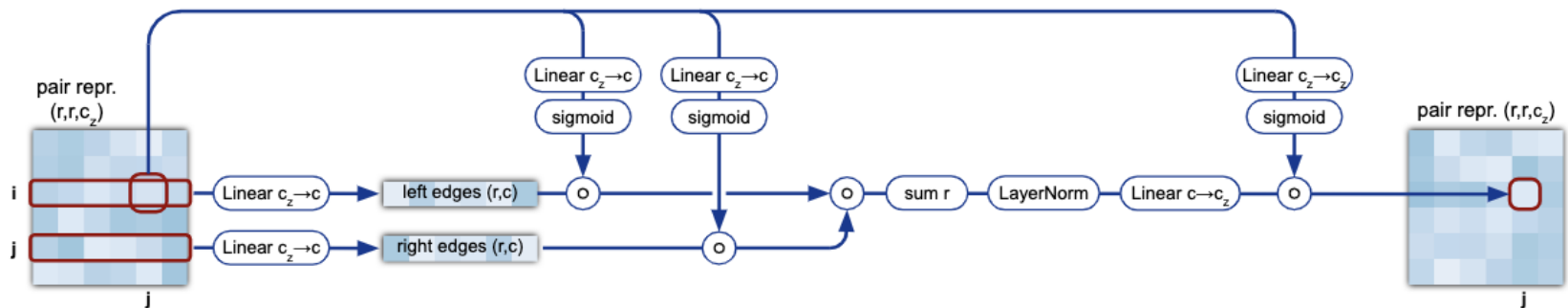
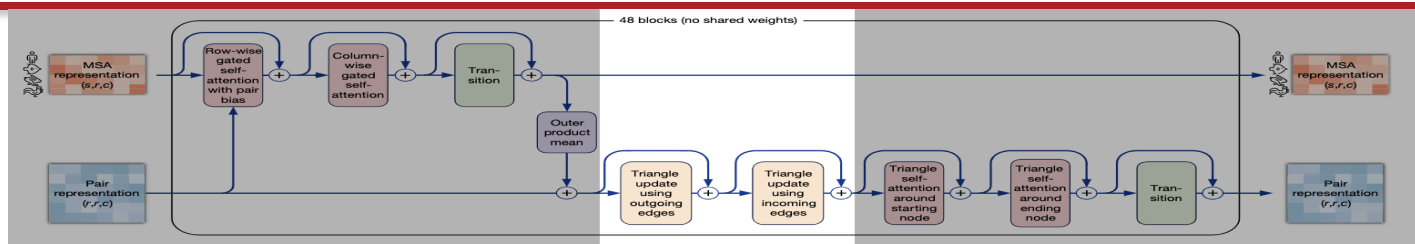


The AlphaFold



Evoformer

Triangle updates

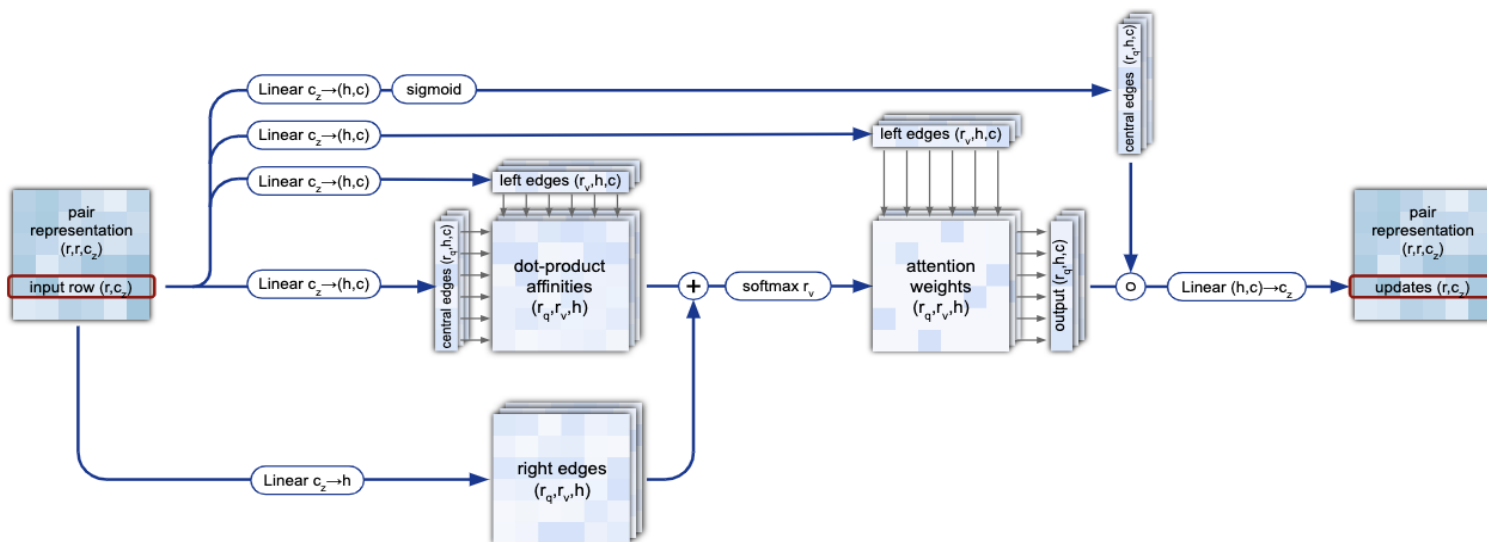
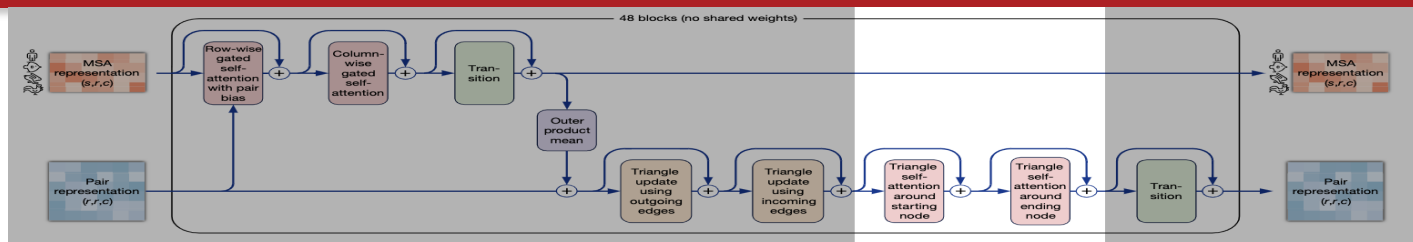


The AlphaFold



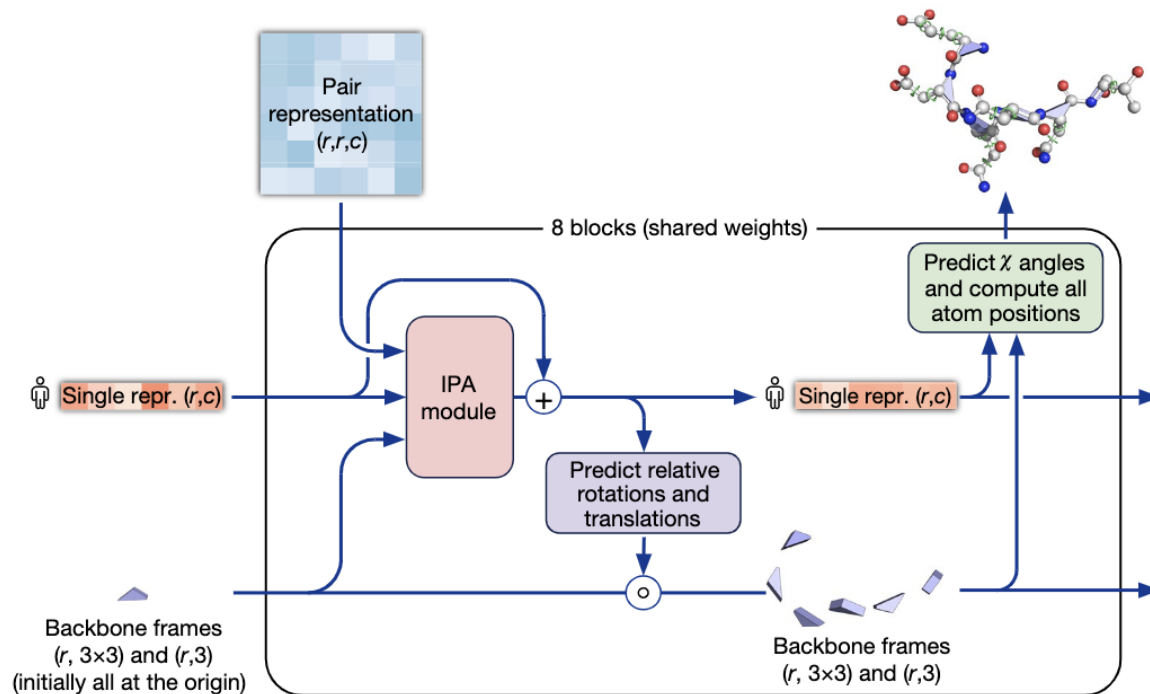
Evoformer

Triangular self-attention





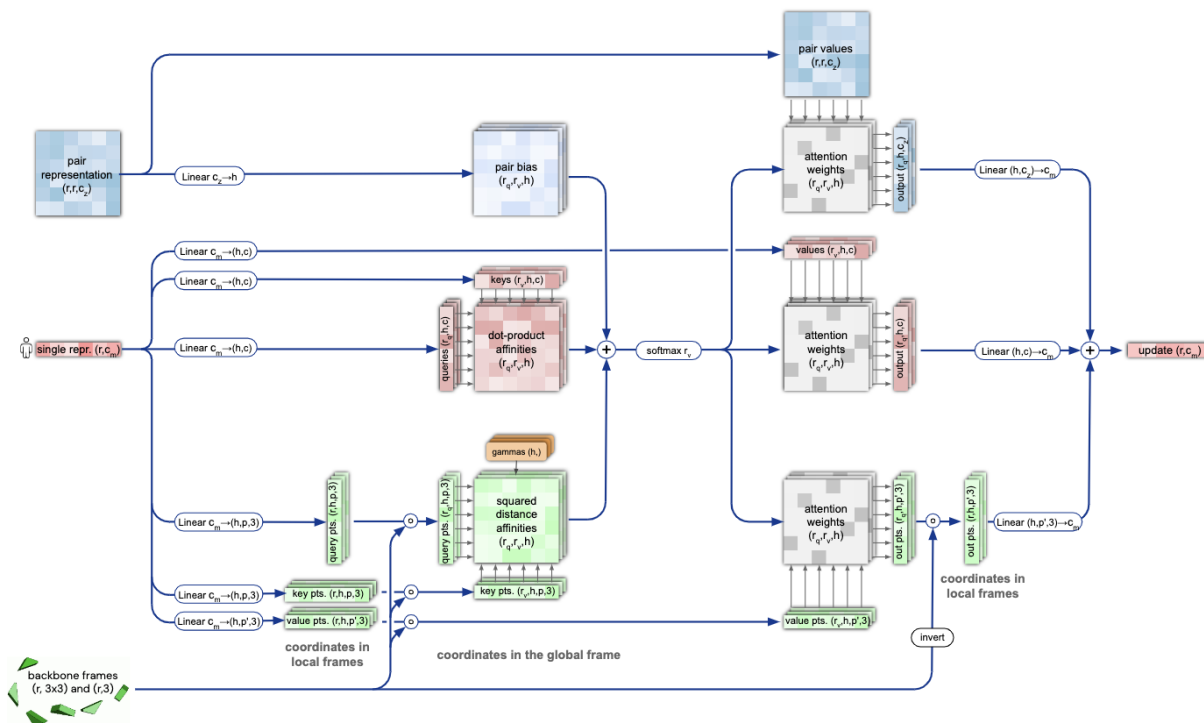
Structure Module



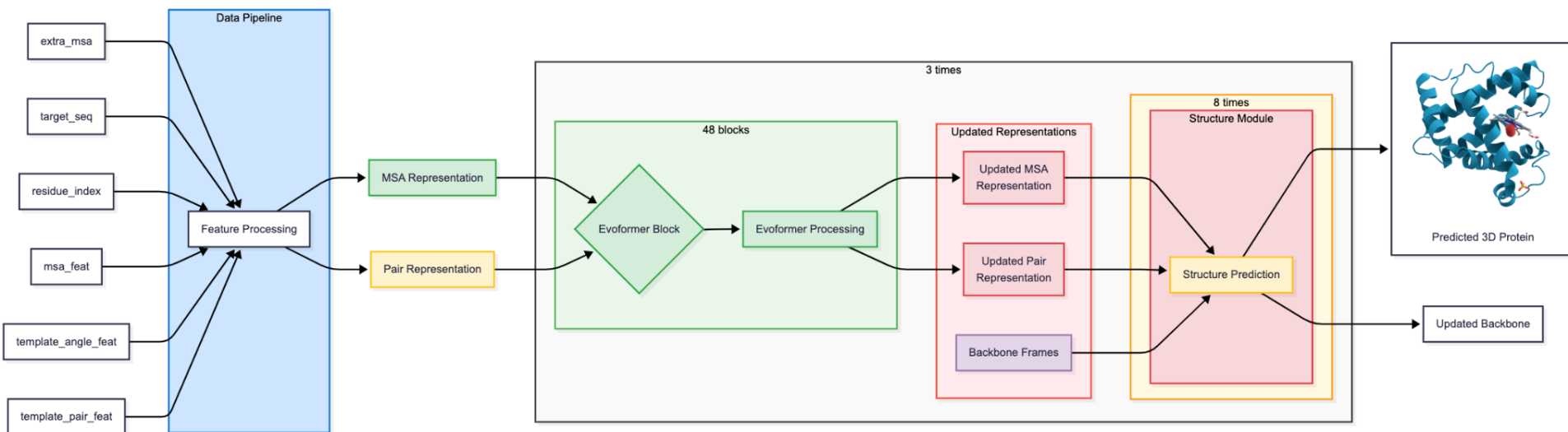


Structure Module

Invariant Point Attention (IPA)



The AlphaFold





Loss function

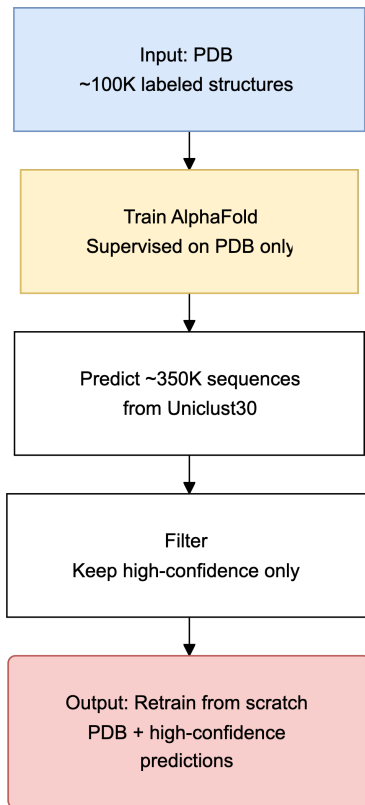
$$\mathcal{L} = \begin{cases} 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} & \text{training} \\ 0.5\mathcal{L}_{\text{FAPE}} + 0.5\mathcal{L}_{\text{aux}} + 0.3\mathcal{L}_{\text{dist}} + 2.0\mathcal{L}_{\text{msa}} + 0.01\mathcal{L}_{\text{conf}} + 0.01\mathcal{L}_{\text{exp resolved}} + 1.0\mathcal{L}_{\text{viol}} & \text{fine-tuning} \end{cases},$$

- $\mathcal{L}_{\text{FAPE}}$ = Frame Aligned Point Error Loss
- $\mathcal{L}_{\text{aux}}$ = is a loss from auxiliary metrics in structure module
- $\mathcal{L}_{\text{dist}}$ = is loss from distogram prediction
- $\mathcal{L}_{\text{msa}}$ = is a loss from BERT-like masked MSA position prediction
- $\mathcal{L}_{\text{conf}}$ = is a loss from model confidence in pLDDT prediction.
- $\mathcal{L}_{\text{exp resolved}}$ = is a loss from a head, predicting, if structure comes from a highly accurate experimental prediction, such as CryoEM or high-resolution X-ray crystallography.
- $\mathcal{L}_{\text{viol}}$ = is a loss for structural violations; penalizes unlikely lengths of bonds, torsion angles and sterically clashing atoms, even if ground truth structure is unavailable.



Self-distillation

Noisy Student Training

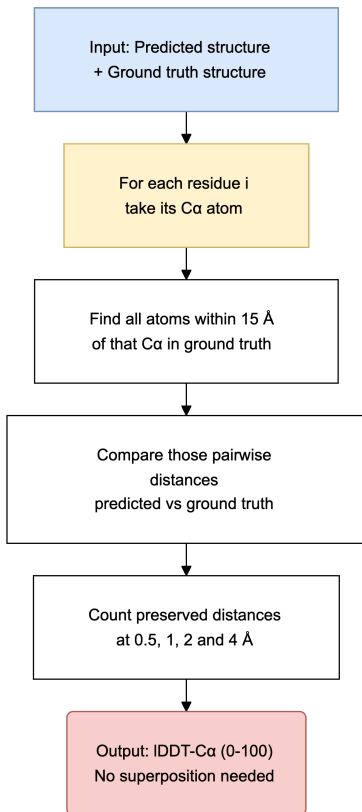


- Labeled data: under 200K structures in the PDB
- Unlabeled data: over 200M known sequences
- Semi-supervised learning, using **Noisy Student Training**
- Cropping and MSA subsampling stop the network simply copying its own predictions
- In the Fig. 4a ablation it is the only change scoring above the baseline



pLDDT

predicted Local Distance
Difference Test



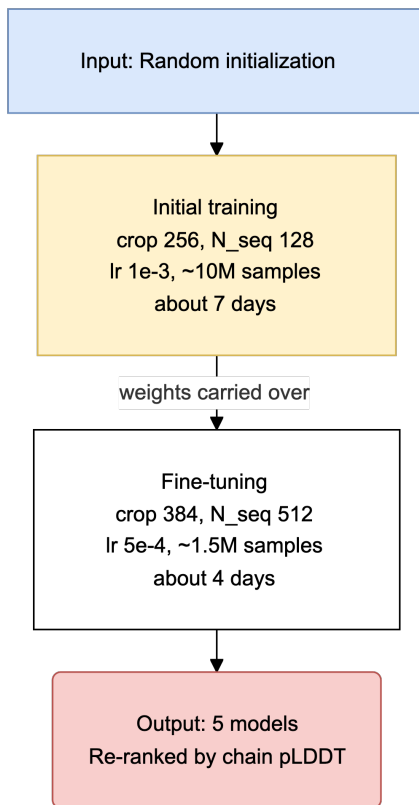
Predicting the score without the answer

- IDDT needs the true structure, so it cannot be computed at prediction time
- pLDDT predicts it from the final single representation s_i
- Projected to 50 bins, each covering 2% of IDDT- $C\alpha$
- Training: cross-entropy against the true bin $\rightarrow L_{\text{conf}}$
- Inference: expected value of the bins $\rightarrow 0-100$
- $\text{IDDT-}C\alpha = 0.997 \times \text{pLDDT} - 1.17$ ($r = 0.76$, $n = 10,795$ chains)
- >90 very high · $70-90$ confident · $50-70$ low · <50 likely disordered



Training protocol

Two stages on 128 TPUv3 cores



- Trained on TPUv3 with a mini-batch of 128, one example per core
- 16 GiB per core is too little for 48 blocks, so activations are recomputed rather than stored
- Violation and “experimentally resolved” losses are switched on for fine-tuning only
- About two weeks end to end, including building the distillation set

**Thank
You!**

